Software Testing Project

Movie Recommendation System

1. Movie Recommendation System Overview

Movie recommendation system is a Java-based system. The system reads movie and user information from two input files, validates the data based on specified rules, and produces a recommendations file listing movie suggestions for each user. If any input error is detected, the output file must contain the first encountered error only.

2. Input Requirements

2.1 Movie Data Requirements

- Movie Title: Every word must start with a capital letter.
- Movie ID: Must consist of all capital letters from the movie title followed by three unique digits.
- Movie Category: One of the allowed categories (e.g., horror, action, drama).

2.2 User Data Requirements

- Username: May contain alphabetic characters and spaces only; must not begin with a space.
- User ID: Must be an alphanumeric string of exactly 9 characters, start with numbers, and may end with at most one alphabetic character. All user IDs must be unique.

3. Input File Format

3.1 movies.txt

- Line 1: movie_title “,” movie_id
- Line 2: category1 “,” category2 ...
- This format repeats for all movies.

3.2 users.txt

- Line 1: username “,” user_id
- Line 2: liked_category1 “,” liked_category2 ...
- This format repeats for all users.

4. Recommendation Logic

For each user: If the user likes category X, the system recommends all movies that belong to category X.

Output format per user:

- Line 1: “For User:” username “,” user_id
- Line 2: “{category1}:” movie_id “-“ movie_title “,” movie_id “-“ movie_title, ...
- Line 3: “{category2}:” movie_id “-“ movie_title “,” movie_id “-“ movie_title, ...
- ...

5. Error Handling

If an error is found in the input files, stop processing immediately and output only the first encountered error. Error messages must follow these formats:

- Movie Title ERROR: {movie_title} is wrong
- Movie Id letters ERROR: {movie_id} are wrong
- Movie Id numbers ERROR: {movie_id} aren't unique
- Username ERROR: {user _name} is wrong
- User Id ERROR: {user_id} is wrong

6. Project Objectives

1. Develop a Java-based movie recommendation system.
1. Analyze system requirements and identify testable functionalities.
2. Develop a comprehensive **Test Plan**.
3. Design **Test Cases** and **Test Scenarios**.
4. Execute manual tests and document results.
5. Automate selected test cases using **Junit**.
6. Generate a final report summarizing findings, defects, coverage, and recommendations

7. Proposed Tools

1. **Test Management:** TestRail
2. **Automation Tools: Junit**
3. **Programming Language:** Java
4. **Bug Tracking:** Jira
5. **Documentation:** PDF

8. Timeline

- March 6th 2026: Developed Java-based movie recommendation system
- March 13th 2026: Test planning
- April 11th 2026: Designed **Test Cases** and **Test Scenarios,** Manual execution and bug reporting
- May 8th 2026: Final submission, that includes the first 2 submissions and Automation scripting, Testing, reporting, refinements

8. Deliverables

1. Test Plan Document
2. Test Cases (as max as possible)
3. Bug Report Log
4. Automation Scripts
5. Test Execution Report
6. Final Project Presentation

Tips:

To speed up building your application, the following links might help you:

1. https://codescracker.com/java/program/java-program-read-file.htm
2. https://codescracker.com/java/program/java-count-occurrence-of-each-word-in-string.htm